Requirement Document: N8N Hands-On Beginner Course

Topic 14: Switch Node & Workflow Routing

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1. Document Control

Field	Value
Document Title	Requirement Document: Topic 14 - Switch Node & Workflow Routing
Course	N8N - Hands On Beginner Course
Target Audience	Trainee / Employee
Version	1.0
Status	Approved
Last Updated	June 25, 2026

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2. Objectives & Scope

2.1 Course Objective

The main objective of this learning module is to teach trainees and employees how to use the Switch node to route items down different paths based on multiple conditions. Trainees will learn how to configure multiple output branches, understand how rules are checked in sequential order, and configure a default branch for items that do not match any rule.

2.2 Learning Scope

- **In Scope:**
  - Adding and configuring the Switch node in the n8n canvas.
  - Configuring the node to read a dynamic value via expression (e.g., `{{ $json.role }}`).

    - Setting up multiple rule branches (e.g., admin, editor, viewer) using string or numeric comparison operators.
    - Setting up a Default fallback branch for non-matching inputs.
    - Connecting different downstream nodes (e.g., Edit Fields nodes) to each output branch to handle items differently.
    - Testing the workflow with multiple items to verify multi-way branching.
- **Out of Scope:**
    - Writing custom JavaScript switch-statement routines inside a Code node (we will focus on the UI configuration of the Switch node).
    - Dynamic generation of output branches based on API response payloads at runtime.

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3. Target Audience Requirements (Pre-requisites)

To successfully complete this hands-on exercise, trainees/employees must have:

- Completed Topic 1 through Topic 13.
- A solid understanding of basic comparison logic (such as string equals) and true/false branching (IF node).

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4. Functional Requirements

Requirement ID	Section	Requirement Description	Priority
REQ-001	Switch Node Integration	The workflow shall include a Switch node to route records along multiple distinct branches based on property matching.	**High**
REQ-002	Multi-Branch Rule Config	The Switch node shall support defining multiple specific comparison rules (e.g., Rule 0: admin, Rule 1: editor, Rule 2: viewer).	**High**
REQ-003	Default Fallback Rule	The Switch node shall support a default/fallback condition routing items to a designated output if no other rules match.	**High**
REQ-004	Sequential Ordering	The Switch node shall evaluate rules in sequential order (top to bottom).	**High**

REQ-005	Dynamic Expression Input	The Switch node value parameter shall dynamically retrieve an input item's property using expressions (e.g., `{{ $json.role }}`).	**High**
REQ-006	Complete Routing Map	The workflow shall route outputs to four distinct downstream Edit Fields nodes matching the roles: • **Set Admin Status** (Branch 0) • **Set Editor Status** (Branch 1) • **Set Viewer Status** (Branch 2) • **Set Guest Status** (Default Fallback Branch 3).	**High**
REQ-007	Multi-Item Processing	The workflow shall process multiple items simultaneously and route each profile to its correct role branch.	**High**

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5. Technical Requirements

ID	Technical Requirement	Details
TR-001	Index-Based Branch Routing	The n8n compiler must translate each rule into an index-based output connector port, routing the item exactly through the port index that matched the rule condition.
TR-002	Dynamic Array Partitioning	The Switch node must process an input array of multiple JSON items and partition them into distinct output arrays corresponding to the matched rules.

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6. Stakeholders and Personas

Persona	Role	Need
Trainee / Employee	Student	Needs to learn how to implement clean multi-path workflows without nesting multiple IF nodes, which is complex and difficult to maintain.

Instructor / Reviewer	Evaluation	Needs to verify that the student correctly configures the Switch node rules, sets the default branch, and links the downstream actions.

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7. Assumptions & Constraints

- **Assumption 1 (ASM-001):** The input mock data contains a field (e.g., `role`) that maps to one of the defined rules or acts as a fallback.
- **Constraint 1 (CON-001):** Once the Switch node output count is set, it is static; adding rules requires updating the node settings and re-connecting the paths.